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Co-designing cybersecurity-related stories with children: Perceptions on cybersecurity risks and parental involvement

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ABSTRACT

Co-designing and co-creating technology have recently received much attention in the child-computer interaction (CCI) community. However, there has yet to be much work to explore the co-designing aspect in the field of children's cybersecurity awareness and parental involvement. With this study, we present the results of a co-design activity with ten children between 9 and 12 years to explore how children think about different cybersecurity issues and parents' roles in those cybersecurity-related situations. Our findings show that children expect various adult roles to provide support to ensure their safety online and especially expect their parents to take a supportive role in cybersecurity experiences by informing them about the risks, offering suggestions on mitigating or protective measures, and helping them take protective measures or actions against any entity posing the risks. Our study contributes to understanding children's perceptions of cybersecurity and parental involvement through storytelling.

1. Introduction

Security is generally an important characteristic of all software products, but the security concern becomes even more crucial when it affects children. Cybersecurity for children has gotten much attention from academia and industry in the last decade, and many applications and platforms have been developed to raise children's awareness of cybersecurity issues (e.g., [39,6]).

However, without direct involvement from the children, it might be challenging to understand or get a clear idea about, for example, the children's mental models, expectations, and thoughts about privacy and security in different online situations and how they think about the role of their parents in their online interactions and risk situations. Research suggests that involving children as co-designers can help researchers understand the best ways to design educational resources that will facilitate children's learning about the complex and diverse concepts related to cybersecurity and privacy [23]. Therefore, we need to put more emphasis on how to include children in the design process of educational resources on cybersecurity.

Moreover, research suggests that we need to design and build technical mediation for children's cybersecurity that is transparent and facilitates the building of trust between children and parents [23]. To facilitate the design and development of such trust-building technical mediation opportunities, we need to understand how children think about the role of their parents and how they expect them to get involved with their cybersecurity experiences. A child's education and learning begin at home, under the guidance of parents, other family members, and caregivers. Children tend to imitate behavior and activities they see people around them doing. Interacting with the parents in learning activities (such as book reading and computer use for education) complements children's cognitive skill development [29,5]. As a result, children's collaboration with other family members and social contact within the family context provides the foundation to observe, learn, and build an understanding of online safety and safe online behavior. Children may find it challenging to learn independently and acquire the necessary skills to understand cybersecurity and risks without enough interaction and guidance from adults since they have different maturity and cognitive abilities than adults. The CCI community and researchers have acknowledged the stakeholder-centric approach to children's technology design, recognizing the influential roles of parents, caregivers, teachers, and peers in children's technology use [28]. These stakeholders, particularly parents and caregivers, have a range of roles to play in directing and controlling children's personal cybersecurity and privacy-related actions and behaviors, whether consciously or unconsciously. We do require parental or caregiver involvement in order to get consent on behalf of children for the use of technical products, to manage internet access and boundaries, and for many other tasks.

With this study, we extend the existing literature on children's

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cybersecurity awareness by investigating children's understanding of different online situations leading to potential cybersecurity risks (i.e., password security, privacy, and online contact) and by contributing to understanding children's perceptions of parental involvement in those cybersecurity situations by doing a storytelling co-design session with children (aged between 9 and 12). Storytelling has received increasing attention in the CCI community as a means to help children express and assign meaning to the world, develop communication, and foster children's ability to interpret, analyze, and synthesize [20]. Thus, in this study, we have used storytelling to understand children's thoughts and address the following research question: *What are children's perceptions about cybersecurity risks, consequences, and parental involvement?* Our findings suggest that children expect their parents to engage in their cybersecurity experiences and learning by taking a supportive role, and though children have knowledge about certain cybersecurity risks and protective measures, they often have a limited comprehension of the implications of the outcomes in the real world and related tradeoffs.

The rest of this paper is organized as follows: Section 2 provides study context and outlines relevant literature. Section 3 discusses our research methodology, and Section 4 presents our results. The findings are discussed in Section 5, and the study finishes with Section 6, which presents the implications of the findings and suggests future research directions.

2. Related work

2.1. Cybersecurity awareness for children

Cybersecurity is an essential aspect of the digital space and a critical learning topic to ensure a safe cyber environment for its users [2]. When spending a lot of time online, all users, regardless of age, are subject to various security threats. Security and privacy threats for children have been a long concern for researchers in the CCI community [33]. Over the past years, concern about children's online etiquette and cybersecurity has increased due to their increasing exposure to technology and various online risks. Password security, exposure to inappropriate content (such as pornographic content, harmful and violent content), cyberbullying, stranger dangers, privacy violations, phishing, etc., are just a few among many others of the common risks the young generation and children frequently face online today [32,36]. Hence, focusing on cybersecurity awareness and developing cybersecurity educational materials for children is a demand of time. Technological development, and so do the security concerns related to it, is a rapidly evolving topic. Therefore, cybersecurity concerns and the development of education resources on cybersecurity need continuous attention from academia and industry.

Given these risks, various multimedia educational tools, such as games, comics, and different training materials, have been designed to improve children's security and privacy knowledge [40]. However, researchers suggest that it is often unclear whether or to what extent children participated in developing the resources targeted for children or whether children find those resources engaging and beneficial for exploring and learning about cybersecurity [27]. Working with children to co-create and co-design educational resources is necessary and beneficial to finding solutions children desire and are interested in using [8]. Therefore, in our study, we involve the children in a co-design session to better understand their expectations and perceptions.

2.2. Parental involvement in children's cybersecurity

Parents use mobile monitoring software and parental controls to observe and restrict their children's online activities to minimize the risks [8,30] and provide a secure digital space for their children. Davis and Koepke [15] suggested that understanding and a strong parent–child relationship can help protect children from online risks more effectively than enforcing parental restrictions. However, research also suggests that children can resent parental control and monitoring applications because they are privacy-invasive, overly restrictive, and harm trust between parents and teens rather than facilitating it [21]. Therefore, understanding the children's perspectives about parental involvement and how the parents can engage in children's cybersecurity management without invading their privacy is crucial. Even though parents might be more interested in collaborating with children in managing cybersecurity than the children [1], emphasizing and reflecting on parent–child collaboration in designing cybersecurity educational applications or parental controls is needed.

However, there has not been enough work that explores and guides the design community on how children expect parental involvement in their on- line and cybersecurity-related affairs beyond parental control and to what extent. Parental involvement is a multidimensional concept that includes 1) parent–child communication, 2) parental supervision, 3) parents' aspirations for children, and 4) parents' active participation [23,18]. It is natural to assume that children will ask for help from their parents in any difficult situations, including for education and learning [22]; children see their parents as role models and trusted partners in helping them assess their capabilities and performance [22]. But in the digital space where parents and children do not necessarily work together or do activities in the same way, the nature of help and support needed from the parents may be different than what is in daily life. To better identify these needs and ways that will be effective and transparent for both parents and children to maintain and manage cybersecurity calls for more attention [7,24,31].

2.3. Co-designing with children

Co-design builds on the tradition of participatory design [35]. According to Burkett [9], co-design is *"the process of designing with people that will use or deliver a product or service"*. Co-design involves collaboration in the design process, reshaping the field of design practice and opening up novel opportunities for collective innovation [34]. During the last decade, co-designing with children has gained immense popularity and has been an important topic in the child-computer interaction community. Co-design is a method that can actively engage several people (such as users or stakeholders) in the design process of a technology or product. Researchers have used co– design strategies in different areas, for example, to understand how children conceptualize intelligent interfaces [38], to design social robots [3], and so on.

In the area of cybersecurity awareness, some researchers have used co- design technique, for example, in designing better parental control applications [30], to better understand children's awareness of online stranger danger [8] and to explore design ideas for privacy-focused games for children [27]. McNally et al. [30] have called for children's inclusion in the parental monitoring software design process to acquire their input on the technology design as it would directly impact and benefit them. Co-design can also contribute to design ideas for cybersecurity and privacy games and educational resources, as demonstrated by Kumar et al. [27]. Therefore, using the co-design technique, we can get a better understanding of children's mental models and how they conceptualize cybersecurity and privacy issues. However, the work on using co-design for designing cybersecurity resources is still limited and needs better inclusion of children in our future endeavors [30].

2.4. Storytelling as a co-design technique

Storytelling is a helpful learning activity for children that can promote imagination and creativity, help develop confidence, and improve memory and sequencing skills [14,19]. Using storytelling and narratives as co-design techniques has been well acknowledged and adapted by CCI researchers (e.g., [14,27]). For example, Kumar et al. [27] conducted storytelling co-design sessions with children to elicit design ideas for new privacy resources. Clegg et al. [10] leveraged a digital storytelling app called StoryKit to help children learn about their life-relevant science experiences and engage in science inquiry through telling stories.

Storytelling as a means helps children to ex- press and enforce their relationships with peers and adults [20]. Researchers have suggested that storytelling and narrative design activities with children can greatly benefit understanding children's imaginations and designing digital games [37]. Therefore, in this study, we use storytelling to explore how children think about different cybersecurity-related online scenarios, imagine the consequences, and whether and how they think about adults (especially their parents) in those stories as elements. The stories made by the children will also contribute to understanding how children might want to design narrative or story-based systems to support cybersecurity learning through games and storytelling.

3. Methods

In this study, we employed the cooperative inquiry (CI) method for the co- design activity. Cooperative inquiry is a set of techniques proposed by Druin [16,17] to work with children as design partners. The cooperative inquiry method can employ a variety of approaches, including participatory design, contextual inquiry, and technology immersion [25,16]. We have employed the contextual inquiry technique in our work, which involves the researchers as well as the participants. We invited the children to create Choose-Your- Own-Adventure (CYOA) stories, following Kumar et al. [27]. In contrast to traditional narratives, which have a fixed storyline, CYOA stories invite the reader to construct the storyline actively. In CYOA stories, the reader makes choices in a given scenario that determine the main character's actions and lead to different outcomes [4,27].

For this study, we gave the children three cybersecurity-related storylines inspired by the existing literature [27,41] during a one-hour design activity. We conducted the activity with pen and paper, where each paper presented one storyline. We asked the children to think about the presented storyline and write a story out of the given context, continuing it as they imagined or wanted. We also welcomed the children to create drawings, if they wanted, in connection to their stories. However, only one of the children drew pictures for the stories (Fig. 1). Each of these storylines initiated a scenario with some details to help the children understand the scenario and guided them with some questions as the decision points to continue the stories, as presented in Table 1.

In addition to understanding the children's thoughts about adults' role in the given cybersecurity scenarios, we also explored how they perceived the associated cybersecurity risks and possible consequences to understand what types of storylines would resonate with children when designing cybersecurity awareness solutions for children in the future. We presented three scenarios to the children but were flexible on how many they needed to finish during the study; we asked them to attempt as many of the scenarios as they wanted (to avoid any pressure that could demotivate the children in the activity).

Table 1
Given cybersecurity-related scenarios for the stories.

ID	Context and decision points
Story 1	One day, Timmy finds a smartphone on the playground. • Should Timmy pick a lost smartphone off the ground and take it home? • Should Timmy check the phone to find out whose phone it is? • Should the phone have a password or not? • Should Timmy go to the house of the phone's owner to return it? • What happens if Timmy picks up the phone?
Story 2	One day, Sara's mom finds out that one of Sara's mobile games sends a lot of information about Sara to other people. Mom wants to remove this game from Sara's phone, but Sara is not sure. • Should Sara listen to Mom and delete the game? • What happens if Sara continues to use the game?
Story 3	One day, Henry goes to a donut shop. There, Henry meets an un- known person. • Should Henry give a checks-in status on social media? • Should Henry give the unknown person his home address (so the person can share a box of donuts with Henry later)? • What happens if Henry does these things?

Fig. 1. Drawing made by a child (C10) for story 1.

3.1. Participant recruitment

A total of 10 children, aged between 9 and 12, participated in this study. We recruited these children through a local children's technology club. We contacted the organizer of the club and invited interested children to participate in our study. In this club, the children practice various activities that relate to technologies, such as making robots using Scratch, building Lego structures, and many other activities. Many of the children also had the experience of participating in robotics competitions and Lego leagues at the national and international levels. This club is a voluntary organization run primarily by the parents of the participating children. The gender distribution of the participant children was three girls and seven boys.

Before conducting the study, we obtained approval from the "XY" Re- search Council (omitted) for data collection and child participation. As children under 13 are not legally permitted to consent, we collected signed in- formed consent from one of the parents for each child. Additionally, before beginning the data collection, we fully disclosed the purpose of the study to the children and their parents, the types of data we would collect from them, and their rights to withdraw from the study at any point.

3.2. Data analysis

We conducted a thematic narrative analysis of the text of stories written by the children [12]. Narrative analysis is an umbrella term for a family of methods that takes stories as the object of inquiries [12,11]. With this analysis, we sought to describe and interpret how the children perceived different themes from the stories. We took a deductive approach [13] to the coding and looking for three categories of themes: perceived roles of adults, perceived risks, and exhibited knowledge about the risks and consequences. We further describe the stories created by the children in relation to the themes in Section 4.

4. Results and findings

As mentioned above in Section 3.2, for each story, we looked at how they pictured the adults' roles in the given situations, how the children conceptualized the situations and perceived the risks, and how they considered the potential outcomes. During the CI session, one of the ten children attempted two stories (skipped story 2), and the rest completed all three stories. Following, we present a summary of our findings in relation to each story.

4.1. Story 1: Finding a smartphone in the playground

Regarding the presence of parents or other adult characters as story elements, we can see that children have mentioned a variety of adult characters in their stories, including parents, police, famous people (like Barack Obama and Casper Lund), as well as fictional characters like Cartman (from the South Park animated series). However, none of the children portrayed these fictional or celebrity characters with any significant engagement or support for the lead characters of the stories. The children visualized these famous people and fictional characters as the owners of the lost phone or the owner had an autograph from the celebrity. In contrast, when parents or other adult figures, such as the

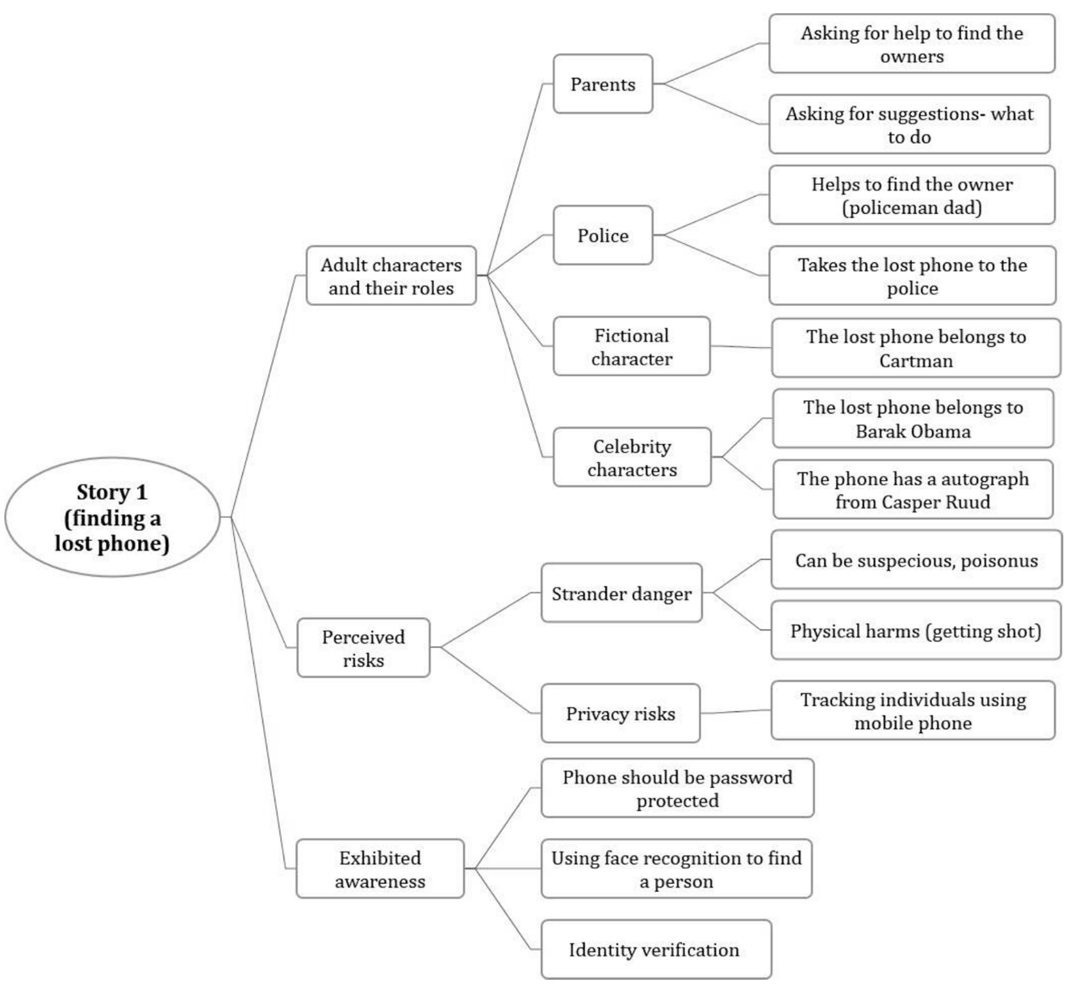

Fig. 2. Themes and related codes from story 1.

police, were portrayed in the stories, the children thought of these characters as someone they could turn to for advice or help. For this particular scenario, we see the children expected their parents to provide support in two ways: first, by guiding them about what they should do in such a situation, and secondly, by helping them to find the owner of the phone. In addition, some of the children expected that the police could contribute to such a scenario by assisting them in finding the owner. In Fig. 2, we present an overview of the themes and the associated codes we identified from children's stories, and in Appendix A.1, we present a summary of the stories the children made out of the given scenario.

In terms of awareness, in this story, children showed a good understanding of the importance of password protection for personal devices. While six of the children explicitly mentioned that the phone should have a password, the rest of the children's stories also indirectly indicated that Timmy (the lead character of the story) could not check inside the phone to find any information about the owner (as the phone was not open). Some children also demonstrated knowledge of additional online security concepts, such as "using face recognition" and "identity verification" when thinking about the privacy protection of personal devices such as mobile phones. Nevertheless, even though the children referenced "using face recognition" and "identity verification" in their stories, they did not associate these concepts with any security issue; instead, they envisioned them as a means to track down the phone's owner (as can be seen in Appendix A.1). Children's perceptions of risks included tracking down a person using a mobile phone tracking app (such as the "Find Me" app on iPhones), physical harm by others (such as Cartman shoots Timmy because Timmy picked up Cartman's phone), and finally, the possibility that an unknown phone could contain something dangerous or suspicious.

4.2. Story 2: A mobile game collecting personal information

In the given scenario, this story already introduces a parent character in the storyline (i.e., Sara's mom). Therefore, we investigated in this narrative how children envisioned this parent role in their story outcomes and whether they considered including any other adult characters except Sara's mother. As presented in Fig. 3, we can see how the children viewed their parents' roles from various perspectives. The children expected their parents to support them by helping them identify and understand the risks, guiding them on how to handle the risks, and taking preventative measures like reporting the game to appropriate authorities. One child also pictured the "game developer" himself as a character in the story, coming to meet Sara and warning her about the dangers of playing a game that would gather a lot of her private details and divulge them to others. Overall, in all these perceived characters and their roles, the children pictured them and their responsibilities as someone who will actively help and guide them to understand the risks and take the right action to prevent or mitigate the consequences. Appendix A.2 presents an overview of all the stories the children made from this scenario.

Regarding cybersecurity awareness, the children showed a good understanding of privacy. All the children concluded that Sara should not play such a game that would collect a lot of her personal information; otherwise, Sara might end up in situations such as getting hacked, scammed, or physically harmed by strangers (for example, being kidnapped). The children also imagined that having access to Sara's personal information could help strangers steal her or her family's

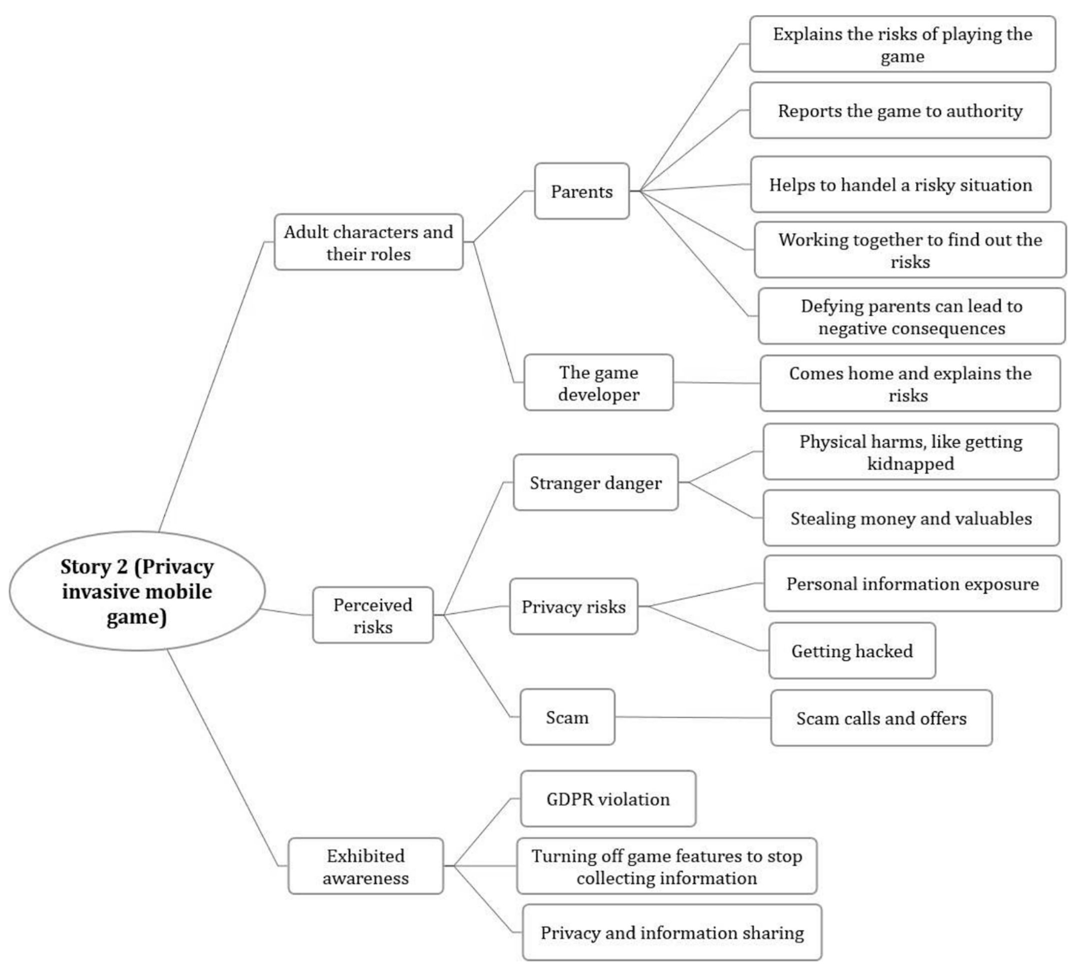

Fig. 3. Themes and related codes from story 2.

valuables and money. In addition to showing awareness of privacy and information sharing and the potential risks associated with privacy violations, two children have brought up topics such as the General Data Protection Regulation (GDPR) and customization of mobile app features to reduce or control user information sharing in their stories. Another child suggested that if a game is gathering user data and breaching their privacy, it should be notified to the appropriate authorities so that they can take appropriate action.

4.3. *Story 3: Meeting a stranger*

This storyline presents a scenario where a character called Henry goes to a donut shop and meets an unknown person. Like in the previous stories, in this scenario also, the children pictured their parents as the people they would turn to for help and advice if they experienced something bad online or needed any suggestion about how to handle that (for example, if someone received weird comments from online strangers). One of the children also mentioned requesting a friend to call the police on his behalf in case he gets kidnapped. Another child also mentioned the police in the story, which pictures Henry being kidnapped and then the police rescuing him. Appendix A.3 presents an overview of the stories the children made for this scenario.

Two of the decision points provided for this scenario were 1) sharing a check-in status online and ii) sharing the home address with an unknown person. Three out of the ten children reflected about sharing a check-in status on social media in their stories; two of them believed it was acceptable to post a status update on social media as long as Henry was cautious about with whom he shared it (i.e., friends on social media or strangers outside of his friend group). And the other child mentioned that such status messages should not be shared online. However, when it came to giving out addresses to strangers, every child made it clear that Henry should not do so. Seven of the participant children continued the given storyline where Henry avoided interacting with the stranger or sharing his personal information. In the stories of the other three children, there were instances where Henry shared his address and then experienced some adverse or risky outcomes, such as losing his valuables, identification (passport), or being kidnapped. It's interesting to note that one of the children who stated Henry had his passport stolen by the stranger in the story also connected this incident of passport theft with the security risk of identity theft. In conclusion, the children could recognize the possible risks of information sharing with strangers from the story's context, and some even showed the ability to reflect on the target audience while posting or sharing information online (Fig. 4).

5. Discussion

5.1. *Children's conceptualization of the risks and consequences*

In the first story, the children demonstrated a good understanding of the need for passwords and password security, but most of the children did not necessarily associate this scenario of picking up a lost phone with any risk. While 9 out of the 10 children imagined the story character Timmy would pick up the phone and try to find the owner, only two children indicated that Timmy should not take the phone and leave it on the ground as it is. However, three of the children imagined that picking up such a lost phone might lead to risky situations, such as being tracked down by strangers using that phone or any kind of physical harm if the phone has something suspicious.

In the second story, the children did show concern in their narratives about privacy violations by the game. They have also imagined other risky situations, such as stranger dangers and becoming victims of hacking and scams led by the breach of privacy in the game. However, not all of the children have concluded this scenario to dire consequences; two children have used it to illustrate more positive learning experiences. In one of these cases, the game developer uses the player's personal data to alert the player about the dangers of playing a privacy-invasive game like this and raise awareness. Another child's story also concluded with a similar positive learning experience, but instead of the game developer helping out, the father steps in to explain why Sara should not play the game and how it can result in privacy violation, leading to other risks as well.

Compared to the previous stories, the children's responses to the third one were more consistent and indicated a better understanding of contacting strangers. Seven (out of ten) children agreed Henry should not talk to strangers or reveal any personal information. When the character, Henry, chooses this option to protect privacy, the story ends.

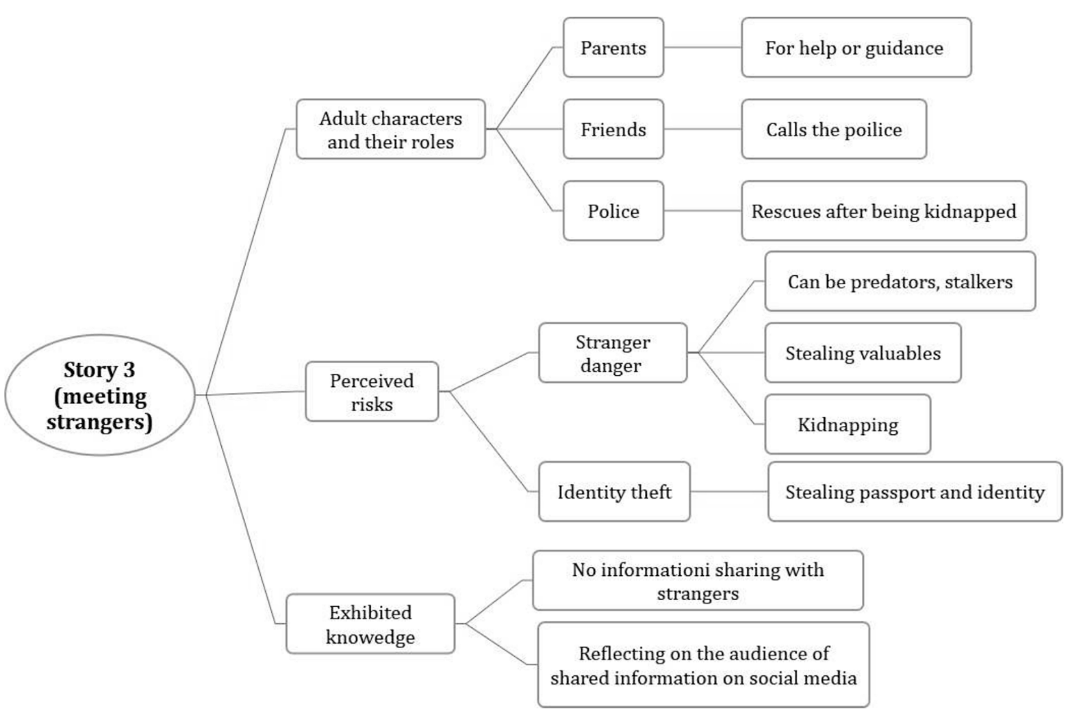

Fig. 4. Themes and related codes from story 3.

Nevertheless, the other five children chose the character to befriend the stranger and share in- formation, and it finally ended with drastic consequences such as kidnapping, theft, and identity theft.

Reflecting on these narratives made by the children, we observe that children show some level of awareness of the given cybersecurity scenarios and relevant protective measures. Although the children did show knowledge of protective measures, they are often limited and lack reflection on the complexity of the actions. Regarding the consequences, we observe that children often see privacy and security concerns more like a black-and-white issue, as also suggested by Kumar et al. [27]. Most children pictured the scenarios with outcomes where they either took safety and privacy-protecting measures or suffered severe consequences. This finding indicates that children often have a limited comprehension of the actions and implications of the outcomes that real-world online security scenarios can bring, as well as the various trade-offs and factors that go into making decisions. We should take this point into account in our future initiatives and adapt cybersecurity awareness materials and education to help children understand the trade-offs and opportunities along with the risks and consequences. Children should realize that using technologies and digital devices not only poses risks or adverse outcomes but also provides ample opportunities for them to learn and benefit.

5.2. *Children's perception about adult's roles*

Our study shows that children visualized different adult characters in their cybersecurity stories with different roles, including their parents, the police, the game developer, and fictional or celebrity characters. We see that where the children imagined the fictional or celebrity characters as just an element of the stories (without much involvement with the lead story character), the other characters had more significant roles. For example, we observed that when the children think about the leading character in a given scenario needing any help or suggestion, they first think about the parents who can help them out. Other than that, some children also thought the police could help them, for example, to find the owner of a lost phone or come to rescue in case of getting kidnapped. For the second scenario we provided, which depicts a privacy-invasive game, one of the children also pictured the game developer in the story who came to the lead character's home to explain how a privacy-invasive game like this can lead to security risks and how to avoid such consequences. It is interesting to see that even though there is a gap in reflecting and ensuring the involvement of different stakeholders in the process of raising cybersecurity awareness of the children from us as researchers [2,31], the children could indeed identify the essential stakeholders that are important and should be involved in ensuring their security in the cyberspace, such as their parents, law enforcement authority and developers of the resources and software for children.

Existing research investigating children's perception of parental controls and monitoring applications shows that children prefer a more collaborative and participatory mediation approach between parents and children rather than the traditional restrictive parental controls [26,30]; children preferred and proposed design concepts for parental controls that can teach them to risk coping and promote parent–child communication and automated interactions. Our findings from this study align with the findings from the previous research. We see from the stories that children expect their parents to play a collaborative and supportive role in their cybersecurity-related activities even when they are not explicitly influenced to do so. This supportive role can be played by the parents, for example, by discussing and explaining the risks to them, providing suggestions on protective or mitigating measures, and also by helping them take formal or legal actions against any entity posing the risks. Nevertheless, when the children did expect help from the parents, we also see how they showed independence and autonomy at the same time. The children expected help and support from their parents instead of depending on their parents to do it for them, as also highlighted by Badillo-Urquiola et al. [8]. Incorporating features that support parent–child collaboration may be more appropriate for children than the traditional paradigm of parental control [7].

5.3. *Implications for design and practice*

The research has potential implications for both designers and practitioners. As we can see from the stories narrated by the children, they have used characters they see or hear about in their everyday lives, such as celebrities or fictional characters from games or animations; this indicates that when designing educational resources for children, using familiar characters and examples that relates to everyday life events can help attract them to the content by facilitating intrinsic motivation and learning. The stories we found from this study can help us understand and design narratives for cybersecurity educational resources for children in the future, which the children will be able to resonate with and connect with.

This storytelling activity with the children can guide us as practitioners on how we can involve the parents and other stakeholders in children's cybersecurity learning, what roles they can play, and how. As we have discussed earlier, children from our study preferred to have autonomy while dealing with online activities and potential risk scenarios; they expected support from the adults in the forms of knowledge and protective actions when needed. While parents can provide such support at home within the family, teachers can also play a similar role in the school environment. Other stakeholders, such as law enforcement authorities and software developers, need to communicate with the people in society more to help them understand how to identify and handle risky situations online, related regulations, and legitimate actions. When the parents have a better insight about these, they are better equipped to teach and help their children.

5.4. *Limitations and future work*

Our study involved a set of ten children who participated in a one-hour storytelling co-design activity and explored three specific themes in this study. Future work can extend upon this study and further investigate children's perceptions of other cybersecurity risk scenarios. Moreover, as the participants in this study were recruited from a technology club, it might be possible that these children have more insights about technology and on- line risks compared to other children who are not involved with such club activities. Future work can draw from a larger, more diverse sample of children with a more balanced distribution of gender to investigate and validate similar work presented in this study.

6. Conclusion

As stated by Kumar et al. [28], *"Privacy and security are multifaceted, context-specific concepts that are challenging to design for"*. With this work, we extend the ongoing efforts to co-design cybersecurity resources with children to meet their needs and expectations better. Our work contributes insights concerning children's perceptions of risks and children's thoughts on parental involvement in their cybersecurity learning and management. We hope to inspire more work in this area of research in co-designing cybersecurity resources by including the children and involving the essential stakeholders, such as parents, in the process. Involving parents in children's security awareness in a way that children will appreciate can make our awareness- raising efforts more efficient and effective. We need to continue investigating and exploring which features and how the implementation of collaborative features can better facilitate collaboration between parents and children to ensure children's online safety rather than create conflict.

CRediT authorship contribution statement

Farzana Quayyum: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Summary of the CYOA stories

Appendix A.1. Story 1: Finding a smartphone in the playground

ID	Age (years)	Outcomes
C1	11	The phone has a password, so he cannot see what is inside. He asks his parents to help him find the owner.
C2	12	Timmy takes the phone to a police officer, who was at the playground at that time. A few moments later, "Barak Obama" comes to Timmy and asks if he saw a phone that he lost in the playground.
C3	11	Timmy takes the phone home and ask his parents what to do. The parents tell Timmy to find the owner and return it back. If Timmy can't figure it out, then to put it back in the playground.
C4	9	Timmy cannot look inside as the phone has a password. Then Timmy finds a boy who says it is his phone. The boy says that his phone has an autograph from Casper Ruud (a professional tennis player) on the back. Timmy returns the phone after verifying the autograph.
C5	11	The phone should have a password. Timmy is unsure what to do; what if the phone is poisoned or suspicious? He asks his mom what to do. Timmy leaves the phone there as Mom tells him to do so. Then, another boy picks up the phone.
C6	11	The phone is locked with a password, so Timmy can not see inside. He decides to take it and ask his parents to help. But rather than going home, he calls his parents to meet him at his school as iPhones can be tracked with the "Find me" app. Later, the parents found a photo on the phone. Using that photo, they find the owner using face recognition.
C7	12	Timmy takes the phone to his policeman dad. Then, Dad finds the owner. Before handing out the phone, Timmy's dad made her tap the phone code (the password) and tell her name and age. After verifying, Dad returns the phone.
C8	11	If this happens in the school, Timmy returns the phone to a teacher or the school administration. Timmy should take the phone home to his parents if this happens outside of school. The phone should have a password, if it does not, they can check if any number is saved for "mamma" or "pappa".
C9	12	Timmy picks up the phone and takes it home. He asks his parents and neighbors if they know whose phone it is. Not finding the owner, Timmy destroys the phone.
C10	12	Timmy and Cartman (a fictional cartoon character) are playing in the playground. Timmy picks up the phone from the ground. Then Cartman shoots Timmy and says, "That's my phone, Timmy!"

Appendix A.2. Story 2: A mobile game and privacy violation

ID	Age (years)	Outcomes
C1	11	Sara should delete the game; otherwise, the whole world will have all the information about Sara.
C2	12	Sara listens to mom and deletes the game. They report the game as it violates the GDPR.
C3	11	Sara listens to Mom and deletes the game. Or, Sara continues playing the game, and a thief steals all their money.
C4	9	Sara continues playing the game and Mom gets upset. But Sara understands when Mom explains to her. Then Sara finds a new game and does not share information with others.
C5	11	Sara should listen to her mom. If she continues playing, she can get hacked. Or, Sara continues playing the game and shares her home location with a stranger from the game. A week later, the stranger comes at Sara's house with a gun. Sara gets scared and hides. It turns out that it was Sara's dad, and he wanted to teach Sara a lesson.
C6	11	N/A
C7	12	Sara does not listen to her Mom and continues to play the game. One day, a man comes to Sara's home and tells her mom that Sara agreed to meet him. But Sara could not understand when and whether she really agreed to meet anyone. Then the man says that he is the creator of the game. He created the game to spot people like Sara, who trusts online people blindly. He warns Sara that she should not blindly trust strangers and every app and website she sees online.
C8	11	Together, they should find out what information is shared and if it can be turned off. They can also search online to determine what information the game collects and if it can be harmful. If this is harmful, they should delete the game.
C9	12	A guy from the game calls and asks if he can come to Sara's home to deliver a PS5, which she has won. Sara brings the phone to her mom and asks what to do. Mom takes the phone and says "No" to the man. Then, Mom deletes the game.
C10	12	Sara's mom tells Sara to delete the game. But Sara doesn't listen and plays the game. She doesn't want to get yelled at, so she doesn't tell Mom. Then Sara gets kidnapped and never returns.

Appendix A.3. Story 3: Meeting a stranger

ID	Age (years)	Outcomes
C1	11	No need to give check-in status. If he shares his home address with the stranger, the stranger will come to Henry's home and steal all his stuff. Henry will become poor.
C2	12	Henry gives his address to the unknown person. The next day, the unknown person visits Henry's home with donuts. A few months later, it turns out that the nice unknown person stole Henry's passport and, therefore, his identity.
C3	11	Henry buys his donut and orders home delivery for his family. But he did not talk to the stranger.
C4	9	Henry is not interested in talking to unknown people. Henry gets his donuts and leaves the shop.

(*continued on next page*)

(*continued*)

ID	Age (years)	Outcomes
C5	11	Henry can give a check-in status on social media. But he should not share his home address with an unknown person. If he shares his address, he can get kidnapped. Or, Henry shares a check-in status and posts a picture (from the donut shop) on Instagram. A minute later, he gets a wired comment that says, "I can't wait to see you." Henry gets scared and runs home from the donut shop. Henry shows the comment to his mom. Mom gets surprised and says, "I didn't know your grandma has an Instagram account!" Henry starts crying and deletes his Instagram account.
C6	11	Henry meets the unknown person, and the person turns out to be very nice. Then they become friends. Later, the person asks Henry for some personal information. Luck- ily, Henry knows a lot about data security and doesn't give any of his personal information away.
C7	12	Henry meets a girl in the donut shop. The girl asks Henry for his social media. But Henry says he does not give personal info to strangers,
C8	11	Henry can give a check-in status to his friends on social media. But if Henry has unknown friends or followers, then he should be careful with this. Henry should not give his home address to the stranger. He does not know if the person has any bad intentions. He could be a stalker or predator.
C9	12	Henry decides not to share any personal information. He buys his own donut and goes home.
C10	12	Henry tells his friend to call the police if he is gone for a while. Then, he is gone for a while, and his friend calls the police. The police find Henry in a basement. So, you should not trust strangers.

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